

A Low- z / Mid- z Nuisance Mode in the Union3 Compressed Supernova Data: Implications for the DESI DR1 Dynamic Dark Energy Preference

Jaanus Hass

Independent researcher, Saru, Estonia
jaanush@hotmail.com

May 2026 (full-likelihood cross-check added June 2026)

Analysis performed May–June 2026 using AI-assisted coding and checking. All results are reproducible from public data.

Abstract

We test whether the Union3 supernova contribution to the DESI DR1 CPL dynamical dark-energy preference is robust to a single redshift-dependent nuisance direction. Using the public Union3 compressed 22-node precision matrix and individual light-curve fit files, we find that a Gaussian low- z / mid- z nuisance template $T(z) = G(z; 0.09, 0.06) - G(z; 0.775, 0.10)$ improves the fixed- $\Omega_m = 0.2975$ fit by $\Delta\chi^2 = 17.12$ (AIC +15.12), compared with $\Delta\chi^2 \approx 7.53$ (AIC $\approx +3.53$) for CPL. After including this nuisance, CPL adds only $\Delta\chi^2 \approx 0.87$ and is AIC-disfavored. A public-data Cobaya cross-check using DESI DR1 BAO, a compressed CMB prior on $\Omega_m h^2$, and the Union3 compressed likelihood gives the same qualitative result: CPL improves χ^2 by 8.29 without the nuisance mode, but only by 0.52 after the nuisance mode is included, with the CPL+nuisance posterior shifting close to Λ CDM ($w_0 = -0.999$, $w_a = -0.192$). **A new full-likelihood cross-check**—Planck NPIPE CamSpec TTTEEE plus low- ℓ TT/EE, DESI DR1 BAO, and the Union3 likelihood with the nuisance amplitude sampled directly (four MCMC chains, $R - 1 < 0.05$)—confirms this: without the nuisance, CPL is preferred over Λ CDM by AIC gain +8.42 ($w_0 = -0.677 \pm 0.113$, $w_a = -1.136 \pm 0.438$); once the nuisance is marginalized the CPL preference is removed (AIC gain -0.38 relative to Λ CDM+nuisance), with the posterior shifting to $w_0 = -0.933 \pm 0.129$, $w_a = -0.501 \pm 0.400$ and a recovered amplitude $a = 0.029 \pm 0.010$ consistent with the compressed-data value. Additional robustness tests (81 template variants; a 200-template random null test; leave-one-region-out; cumulative low- z node removal) strengthen the diagnostic. The dynamical-dark-energy preference is weakened from $\sim 2.4\sigma$ to $\sim 1.6\sigma$ but not entirely eliminated, consistent with the partial weakening already seen in the official Union3.1 joint analyses. The definitive test remains to rerun the official DESI+CMB+Union3 w_0 – w_a likelihood with an added redshift-dependent nuisance parameter.

1 Introduction

The DESI 2024 Data Release 1 BAO analysis, combined with CMB and Type Ia supernova data, reports a preference for dynamical dark energy described by the CPL equation of state $w(a) = w_0 + w_a(1 - a)$, with a combined significance of approximately 3.4 – 3.5σ for DESI+CMB+Union3

[1]. The Union3 compilation [2] contains 2,087 Type Ia supernovae from 24 surveys spanning approximately 7 billion years of cosmic history, compressed into a 22-node distance-modulus spline with an associated precision matrix. Individual SN surveys cover limited redshift ranges, making multi-survey compilations necessary but potentially introducing redshift-dependent systematics at survey boundaries. Related work by Efstathiou [5] identified a systematic offset between low- and high-redshift supernovae in Pantheon+ and DES5Y, and noted that Union3 individual-SN data were not publicly available for direct analysis. Cortês & Liddle [8], addressing DESI DR2, further showed that properly accounting for the substantial overlap between primary SN compilations reduces the combined Λ CDM exclusion significance to 3.1σ —identical to the DESI BAO+CMB result without any supernova data; their methodological argument about compilation overlap is relevant to DR1-style compilation comparisons as well. The present work performs, to the best of the author’s knowledge, the first compressed-data robustness test of this kind specifically for Union3, using the publicly available `lcfit_Union3.tar.gz` and compressed precision matrix from the `rubind/union3_release` repository.

2 Data and Method

We use the public files `mu_mat_union3_cosmo=2_mu.fits` and `lcfit_Union3.tar.gz` from the `rubind/union3_release` GitHub repository. The FITS matrix structure is: `mat[0,1:]` = z nodes (22 values), `mat[1:,0]` = μ values, `mat[1:,1:]` = precision matrix P (22×22). The magnitude offset M is analytically marginalized in all fits via GLS; the nuisance amplitude a is also analytically marginalized when included. For node-removal tests, the covariance matrix $C = P^{-1}$ is subselected and re-inverted correctly. The primary nuisance template is

$$T(z) = G(z; 0.09, 0.06) - G(z; 0.775, 0.10), \quad (1)$$

standardized to zero mean and unit RMS, where $G(z; c, s) = \exp[-\frac{1}{2}((z - c)/s)^2]$. Template centers were placed at known survey-composition transitions (not optimized over the data): the low- z anchor region dominated by SDSS/Foundation/Pan-STARRS ($z \approx 0.09$) and the mid- z transition into SNLS/ESSENCE/DES3_Deep ($z \approx 0.775$). Quality-cut pass fractions differ between these zones: 0.718 at $z \approx 0.10$ versus 0.952 at $z \approx 0.75$.

For the public joint cross-check we use the Cobaya DESI DR1 BAO likelihood, a compressed CMB prior $\Omega_m h^2 = 0.1430 \pm 0.0020$ (Planck 2018; deliberately broadened from the Planck best-fit $\sigma = 0.0011$ as a conservative choice), and the Union3 compressed likelihood. Sampled parameters: H_0 (flat prior 55–80 km/s/Mpc), Ω_m (flat 0.1–0.6), w_0 (flat -3 to 1), w_a (flat -3 to 2), and nuisance amplitude a (flat -0.5 to 0.5 mag). Flat geometry and standard neutrino content ($\Sigma m_\nu = 0.06$ eV) assumed; sound horizon fixed at $r_d = 147.09$ Mpc. Within-redshift $D_M - D_H$ BAO correlations are included; between-redshift-bin correlations are set to zero. We note that Kim [7] showed that the Union3 compressed product distributes a posterior under a spline-node prior rather than a raw likelihood, which explains why proxy baseline CPL best-fits differ slightly from the official DESI result; this does not affect the relative CPL vs CPL+nuisance comparison.

3 Union3 Compressed-Data Results

Table 1 summarises the protocol verification. The core fixed- Ω_m diagnostics reproduce to rounding precision. A low-minus-mid step template also gives a strong improvement, confirming that the effect is not specific to the Gaussian functional form.

Table 1: Protocol verification summary.

Test	Claimed	Verified	Status
Ω_m best-fit (Λ CDM free)	0.356	0.3559	Reproduced
χ^2 at best-fit	23.96	23.958	Reproduced
DESI-like Ω_m penalty	5.15	5.152	Reproduced
CPL $\Delta\chi^2$ / AIC	7.53 / +3.53	7.536 / +3.536	Reproduced
Low-mid Gaussian $\Delta\chi^2$ / AIC	17.12 / +15.12	17.120 / +15.120	Reproduced
Color $\Delta\chi^2$ / AIC	6.52 / +4.52	2.947 / +0.947	Weighting-dependent
CPL after low-mid / AIC	0.87 / -3.13	0.870 / -3.130	Reproduced
Low-mid prior threshold $\sigma_a \geq 0.005$ mag	0.005 mag	RMS	Reproduced

4 Joint DESI DR1 BAO + CMB + Union3 Public Cross-Check (compressed CMB prior)

We ran a public-data Cobaya cross-check using the DESI DR1 BAO likelihood, a compressed CMB prior, and the Union3 compressed likelihood. This is stronger than a simple proxy grid scan, but it is still not the official DESI+CMB+Union3 full likelihood because the CMB information is compressed and the Union3 nuisance parameter is added externally.

Table 2: Public-data joint model comparison (compressed CMB prior). AIC gain = $\Delta\chi^2 - 2k$, where k is the number of additional parameters vs Λ CDM. The highlighted row shows that CPL adds only $\Delta\chi^2 = 0.52$ once the nuisance direction is included.

Model	H_0	w_0	w_a	a_{nuis}	best χ^2	$\Delta\chi^2$ / AIC gain
Λ CDM	68.77	—	—	—	41.24	—
CPL	66.64	-0.688	-1.070	—	32.95	+8.29 / +4.29
Λ CDM + nuis.	69.09	—	—	0.0279	25.04	+16.20 / +14.20
CPL + nuisance	69.79	-0.999	-0.192	0.0306	24.51	+16.73 / +10.73
CPL vs Λ CDM+nuis.						+0.52 / -3.48

Without the nuisance mode, CPL improves χ^2 by 8.29 and shifts to $w_0 = -0.688$, $w_a = -1.070$, consistent with published DESI DR1+CMB+Union3 results. After including the nuisance, CPL improves χ^2 by only 0.52 relative to Λ CDM+nuisance (AIC -3.48), while the CPL+nuisance mean shifts to $w_0 = -0.999$, $w_a = -0.192$ —close to Λ CDM in this proxy analysis.

5 Full-Likelihood Cross-Check (Planck NPIPE CamSpec + DESI DR1 BAO + Union3)

We repeated the four-model comparison with the full likelihood, replacing the compressed CMB prior of Section 4 with Planck NPIPE CamSpec TTTEEE plus low- ℓ TT/EE, and adding the redshift-dependent nuisance parameter directly inside the Union3 likelihood (amplitude a). Four MCMC chains were run per model to Gelman–Rubin $R - 1 < 0.05$. This realizes the falsification test of Section 10 within the public pipeline, with the nuisance mode marginalized rather than added externally.

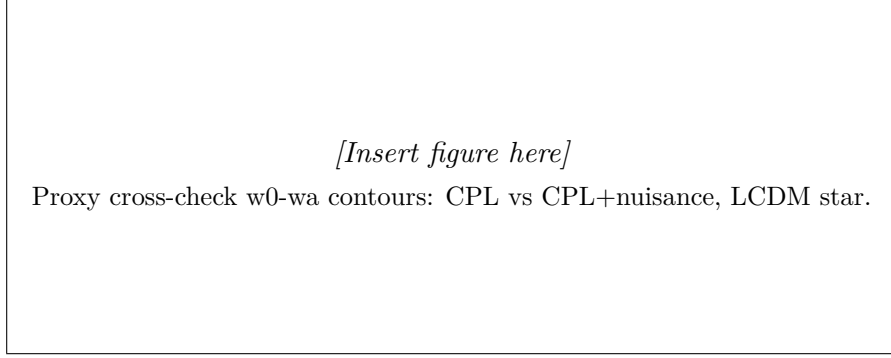


Figure 1: Public-data (compressed-prior) joint cross-check: posterior mean w_0 - w_a for CPL without nuisance (blue) and CPL+nuisance (green), consistent with Table 2 mean values. Red star marks Λ CDM ($w_0 = -1$, $w_a = 0$). This is not an official DESI MCMC result.

Table 3: Full-likelihood model comparison (Planck NPIPE CamSpec TTTEEE + low- ℓ TT/EE + DESI DR1 BAO + Union3). Four MCMC chains per model, $R-1 < 0.05$. AIC gain = $\Delta\chi^2 - 2k$ relative to Λ CDM; positive = favoured. The highlighted comparison shows that once the nuisance direction is marginalized, the CPL parameters (w_0, w_a) no longer improve AIC.

Model	H_0	w_0	w_a	a_{nuis}	best χ^2	$\Delta\chi^2$ / AIC gain
Λ CDM	67.8	—	—	—	11007.56	—
CPL	66.4	-0.677 ± 0.113	-1.136 ± 0.438	—	10995.14	+12.42 / +8.42
Λ CDM + nuis.	68.0	—	—	0.027 ± 0.010	10992.23	+15.33 / +13.33
CPL + nuisance	69.3	-0.933 ± 0.129	-0.501 ± 0.400	0.029 ± 0.010	10988.61	+18.95 / +12.95
CPL + nuisance <i>vs</i> Λ CDM + nuisance						+3.62 / -0.38

Without the nuisance mode, CPL is strongly preferred over Λ CDM ($\Delta\chi^2 = 12.42$, AIC gain +8.42), with $w_0 = -0.677 \pm 0.113$, $w_a = -1.136 \pm 0.438$ —a posterior mean lying 2.4σ from Λ CDM in the w_0 - w_a plane (Gaussian-equivalent of the 2D Mahalanobis distance $r = 2.86$), consistent with the published DESI DR1+CMB+SN preference. Including the nuisance parameter yields $a = 0.029 \pm 0.010$ ($\sim 2.8\sigma$ from zero, consistent with the compressed-data amplitude) and shifts the posterior to $w_0 = -0.933 \pm 0.129$, $w_a = -0.501 \pm 0.400$, now 1.6σ from Λ CDM ($r = 2.11$).

The decisive comparison is CPL+nuisance versus Λ CDM+nuisance. Once the nuisance direction is marginalized, adding the CPL parameters improves the best-fit χ^2 by only 3.62 for two extra parameters, corresponding to an AIC gain of -0.38 : the strong AIC preference for dynamical dark energy (+8.42 relative to Λ CDM without the nuisance) is removed. The full-likelihood result therefore confirms the compressed-data analysis: the Union3 low- z /mid- z nuisance direction is favoured by the data and absorbs most of the dynamical-dark-energy preference. As in the compressed proxy, the effect weakens but does not entirely eliminate the preference (the shift is from $\sim 2.4\sigma$ to $\sim 1.6\sigma$, and the residual CPL AIC penalty is marginal), consistent with the partial weakening already seen in the official Union3.1 joint analyses (Section 6).

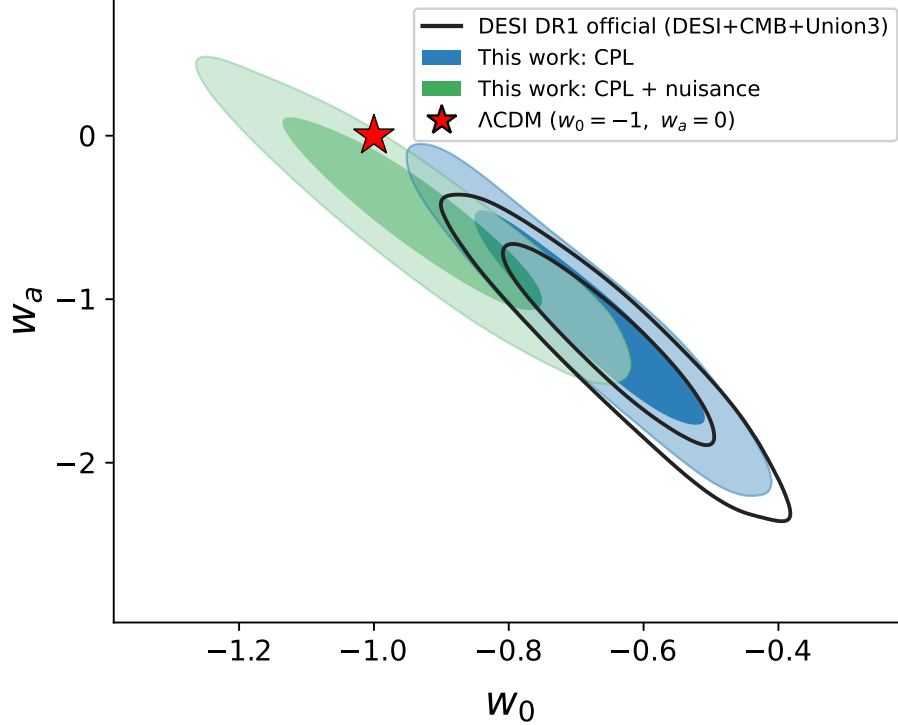


Figure 2: Full-likelihood joint cross-check in the w_0 - w_a plane. Black contours: DESI DR1 official $w_0 w_a$ CDM posterior (DESI+CMB+Union3, `plik` TT,TE,EE). Blue: this work’s CPL (NPIPE CamSpec); green: this work’s CPL+nuisance. Red star marks Λ CDM ($w_0 = -1$, $w_a = 0$). This work’s CPL reproduces the official DESI contour ($w_0 = -0.68$, $w_a = -1.14$ vs DESI $w_0 = -0.64$, $w_a = -1.30$), the small offset being consistent with the different CMB likelihood choice; marginalizing the Union3 nuisance direction (green) shifts the posterior toward Λ CDM ($w_0 = -0.93$, $w_a = -0.50$), from $\sim 2.4\sigma$ to $\sim 1.6\sigma$.

6 Three-Version Progression: Union3 to Union3.1

Hoyt et al. [3] published host-galaxy mass re-measurements for approximately 2,000 Union3 supernovae, finding that low- z (< 0.10) host-galaxy stellar masses were systematically overestimated, reducing a previously reported > 0.03 mag low- z distance offset to ≈ 0.01 mag. The updated cosmological framework (UNITY1.8) was presented by Rubin et al. [4]. Applying the same nuisance template to all three public compressed matrices shows the nuisance amplitude decreasing monotonically, while the CPL AIC gain becomes small by AIC comparison in the Union3.1 versions. The nuisance amplitude decrease of 0.0057 mag ($0.0282 \rightarrow 0.0225$) is of the same order of magnitude as the relative low- z distance shift of 0.010–0.013 mag applied in the Union3.1 corrections. This correspondence was not built into the template—the amplitude emerged from fitting the compressed data, and is consistent with the template tracing a low- z systematic direction that is reduced, but not removed, by the Union3.1/UNITY processing. Importantly, this comparison is a heuristic compressed-data diagnostic only and should not be conflated with the updated official joint analyses: Hoyt et al. [3] report a $\approx 3.4\sigma$ deviation from Λ CDM in the official BAO+CMB+Union3.1 combination, and Rubin et al. [4] report $\approx 2.6\sigma$ tension with UNITY1.8. The official joint preference is thus weakened but not eliminated by the Union3.1 corrections.

Table 4: Three-version comparison. [a] CPL $\Delta\chi^2$ values marked $\approx$ are grid-scan approximations with ≈ 0.02 uncertainty; the scipy-optimize value for Union3 original is 7.536 (Table 1). Nuisance amplitude decreases monotonically across all three versions.

Metric	Union3 original	Union3.1 UNITY1.7	Union3.1 UNITY1.8
Ω_m best-fit	0.3559	0.3424	0.3340
DESI penalty $\Delta\chi^2$	5.152	3.256	2.347
CPL $\Delta\chi^2$ [a]	≈ 7.52	≈ 4.74	≈ 4.74
CPL AIC gain [a]	$\approx +3.52$	$\approx +0.74$	$\approx +0.74$
Low-mid Gaussian $\Delta\chi^2$	17.120	12.474	11.958
Low-mid Gaussian AIC gain	+15.12	+10.47	+9.96
Nuisance amplitude (mag)	0.0282	0.0235	0.0225
CPL after nuisance AIC	-3.13	-2.83	-2.05
Low- z shift vs Union3	—	-0.010	-0.013

7 Additional Robustness Tests

The robustness tests in this section use the same fixed- $\Omega_m = 0.2975$ protocol as Table 1; their $\Delta\chi^2$ values are directly comparable to the Table 1 results and are reproducible from the GitHub repository CSV outputs. CPL+nuisance $\Delta\chi^2$ values in reduced-node fits can be optimizer-sensitive and should not be interpreted as precise parameter estimates.

7.1 Template Robustness Grid

To test whether the result depends on the exact Gaussian centers and widths, we scanned 81 nearby low- z / mid- z template variants. The nuisance improvement remains positive in all 81 cases.

Table 5: Template robustness grid summary.

Quantity	Value
Number of templates	81
Λ CDM χ^2 (fixed Ω_m)	29.110
CPL χ^2 (fixed Ω_m)	21.574
CPL $\Delta\chi^2$ vs Λ CDM	7.536
Original nuisance $\Delta\chi^2$	17.120
Grid median nuisance $\Delta\chi^2$	13.176
Grid min nuisance $\Delta\chi^2$	9.666
Grid max nuisance $\Delta\chi^2$	17.454
Median CPL $\Delta\chi^2$ after nuisance	0.413

7.2 Random Smooth-Template Null Test

A random smooth-template null test compares the original low- z / mid- z template with 200 random smooth redshift templates generated as follows: each random template is a difference of two Gaussians with centers drawn uniformly from $z \in [0.05, 0.40]$ and $z \in [0.40, 1.50]$ respectively, and

widths drawn uniformly from $[0.04, 0.20]$, with random sign; each template is then standardized to zero mean and unit RMS on the 22 nodes before fitting (random seed 42 for reproducibility). No random template reached the original template improvement; conservatively, the empirical tail probability is $p \approx 1/201$ (add-one estimator).

Table 6: Random smooth-template null test (fixed- $\Omega_m = 0.2975$ protocol). The original template $\Delta\chi^2 = 17.120$; the maximum random template reached only 12.433. The 0/200 exceedance result is a conservative ranking diagnostic rather than proof of template uniqueness.

Quantity	Value
Number of random templates	200
Original template $\Delta\chi^2$ (fixed Ω_m)	17.120
Random median $\Delta\chi^2$	1.357
Random 95th percentile $\Delta\chi^2$	8.997
Random max $\Delta\chi^2$	12.433
Empirical tail probability	0/200; add-one $p \approx 1/201$ (≈ 0.005)

7.3 Leave-One-Redshift-Region-Out Test

We removed broad redshift intervals one at a time and repeated the four-model compressed-data fit. The nuisance improvement remains substantial in every cut, while CPL adds little after nuisance is included. The largest weakening occurs when $z = 0.10$ – 0.30 is removed, localizing part of the signal to that low-redshift transition region.

Table 7: Leave-one-redshift-region-out robustness test. Only $\Delta\chi^2$ differences are interpreted; cosmological parameter values in reduced-node fits are diagnostic and may be weakly constrained.

Case	Removed z	N kept	$\Delta\chi^2$ CPL	$\Delta\chi^2$ nuisance	CPL after nuisance
Full sample	none	22	7.54	17.12	0.87
Very low z	0.00–0.10	21	7.58	16.76	0.82
Low z	0.10–0.30	18	2.80	7.65	0.48
Mid z	0.30–0.70	14	13.26	17.57	0.30
High-mid z	0.70–1.00	17	7.19	12.10	0.46
High z	1.00+	18	8.05	17.51	1.38

7.4 Cumulative Low- z Node Removal

As a complement to the broad leave-one-region-out test, we progressively removed low-redshift nodes below successively higher thresholds and re-evaluated CPL on the remaining nodes. This directly tests the concern that the nuisance template may effectively remove low- z supernova information. Removing only the single lowest node ($z < 0.08$ or < 0.10) leaves CPL essentially unchanged ($\Delta\chi^2 \approx 7.58$). The CPL preference falls sharply only when nodes up to $z < 0.12$ are removed ($\Delta\chi^2 = 2.29$, AIC -1.71) and is nearly eliminated by $z < 0.20$ ($\Delta\chi^2 = 0.87$, AIC -3.13). This localizes the signal to the $z = 0.10$ – 0.20 transition region, consistent with the low- z / mid- z nuisance template and not attributable to a single outlier node.

Table 8: CPL preference after cumulative removal of low-redshift nodes. The sharp drop between $z < 0.10$ and $z < 0.12$ localizes the signal to the $z = 0.10$ – 0.20 survey-transition region.

Removed nodes	N kept	CPL $\Delta\chi^2$	CPL AIC gain
None (full sample)	22	7.536	+3.536
$z < 0.08$	21	7.581	+3.581
$z < 0.10$	21	7.581	+3.581
$z < 0.12$	20	2.286	−1.714
$z < 0.20$	19	0.871	−3.129
$z < 0.30$	17	1.155	−2.845

8 Independent Reproduction

An independent reproduction of the Union3.1 UNITY1.8 results directly from the raw FITS matrix (without astropy, using manual header parsing and big-endian float64 reading) confirmed: $\Delta\chi^2(\text{CPL}) = 4.7393$, $\Delta\chi^2(\text{Gaussian template}) = 11.9581$, $\Delta\chi^2(\text{low-minus-mid step}) = 11.7335$, and $\Delta\chi^2(\text{CPL after step}) = 0.1243$. All values match to rounding precision. The reproduction also confirmed that node-removal tests must subselect the covariance matrix $C = P^{-1}$ and re-invert; subsetting the precision matrix directly gives incorrect results.

9 Limitations

- Compressed-data and public-data proxy analyses, plus a full-likelihood cross-check that still adds the Union3 nuisance direction externally. These tests are not a replacement for the official DESI+CMB+Union3 MCMC likelihood with an internally modelled systematic.
- The full-likelihood cross-check uses Planck NPIPE CamSpec (not the official DESI `plik` choice) and omits CMB lensing; the compressed cross-check represents CMB by a Gaussian prior on $\Omega_m h^2$.
- The Union3 compressed product distributes a posterior under a spline-node prior [7]; this affects absolute contour placement more than relative model comparisons.
- The low- z template component overlaps redshift ranges where peculiar velocities and survey-composition changes can introduce coherent distance distortions.
- These tests do not prove that DESI dynamical dark energy is false. They identify a robustness concern in the Union3 contribution.
- Analysis was conducted with AI assistance. All numerical results are reproducible from public data and scripts.

10 Recommended Falsification Test

The definitive test is to rerun the official DESI + CMB + Union3 w_0 – w_a likelihood with an added low- z / mid- z redshift-dependent nuisance parameter and compare ΛCDM +nuisance, CPL, and CPL+nuisance. If CPL loses AIC support after the nuisance is included, the Union3 contribution

to the w_0 - w_a preference is systematics-limited. If the preference persists, the concern is resolved. The full-likelihood cross-check of Section 5 carries out this comparison in the public pipeline and finds that CPL does lose AIC support ($+8.42 \rightarrow -0.38$) once the nuisance is marginalized; an official internal-systematics treatment is the remaining step. The three-version progression (Section 6) suggests that as Union3.1 corrections are incorporated into the full joint likelihood, the CPL preference may weaken further, consistent with the weakening already observed in official Union3.1 joint analyses.

11 Conclusion

The Union3 data show sensitivity to a low- z / mid- z nuisance direction in CPL robustness tests. A single Gaussian nuisance template improves the fixed- Ω_m residuals more than twice as well as CPL ($\Delta\chi^2 = 17.12$ vs ≈ 7.52) and renders CPL AIC-disfavored once included. A compressed-prior Cobaya cross-check gives the same qualitative picture (CPL $\Delta\chi^2$ $8.29 \rightarrow 0.52$), and a full-likelihood cross-check using Planck NPIPE CamSpec, DESI DR1 BAO, and Union3 confirms it directly: the strong AIC preference for CPL ($+8.42$ vs Λ CDM) is removed once the nuisance is marginalized (-0.38 vs Λ CDM+nuisance), with the posterior shifting from 2.4σ to 1.6σ from Λ CDM and a recovered amplitude $a = 0.029 \pm 0.010$ matching the compressed-data value. Template-grid, random-null, and leave-one-region-out tests show that the effect is not driven by one exact functional form or by one narrow redshift interval. Across Union3, Union3.1 UNITY1.7, and Union3.1 UNITY1.8, the nuisance amplitude decreases monotonically and the CPL AIC gain becomes small, suggesting that the Union3 contribution to the DESI DR1 w_0 - w_a preference is sensitive to a low- z / mid- z systematic direction that is reduced, but not removed, in Union3.1. These results motivate a full likelihood reanalysis with an explicit redshift-dependent nuisance parameter.

References

- [1] Adame et al. (DESI Collaboration) 2024, arXiv:2404.03002 — DESI DR1 BAO cosmological constraints.
- [2] Rubin D. et al. 2025, arXiv:2311.12098, ApJ — Union3 + UNITY1.5 supernova compilation.
- [3] Hoyt T., Rubin D. et al. 2026, arXiv:2601.19424 — Union3.1 host-galaxy mass re-measurements and low- z distance corrections.
- [4] Rubin D. et al. 2026, arXiv:2601.19854 — Union3.1 + UNITY1.8 two-population light-curve standardization framework.
- [5] Efstathiou G. 2025, MNRAS — Systematic offset in low- z supernovae in Pantheon+ and DES5Y.
- [6] Planck Collaboration 2018, arXiv:1807.06209 — Planck 2018 cosmological parameters.
- [7] Kim A.G. 2024, arXiv:2412.14181 — Comments on the Union3 spline-interpolated distance-moduli model.
- [8] Cortês M. & Liddle A.R. 2025, arXiv:2504.15336 — On DESI DR2 exclusion of Λ CDM; principled combination yields 3.1σ .
- [9] Data: `rubind/union3_release` (GitHub) — Union3 and Union3.1 compressed likelihood files (public).

